

Supplementary Information

Background matching audits mass spectral antimicrobial resistance prediction

Supplementary Note 1: Why background matching is needed

Raw external AUC answers whether a model ranks resistant isolates above susceptible isolates at a target site. It does not answer whether that ranking is driven by focal-drug biology or by the resistant population in which the label occurs. In bacterial clinical data, resistance labels are structured by lineage, mobile elements, local prescribing ecology and co-resistance. A MALDI-TOF model can exploit this structure because spectra encode abundant cellular proteins and lineage-associated peak patterns.

The background-matched audit controls the observable co-resistance component of this structure. For each focal drug, isolates are grouped by the AST labels of the other drugs in the same panel. The focal drug is excluded from the signature. Within valid strata, resistant and susceptible isolates have the same observed co-resistance background. If the model still separates them, it has residual within-background signal. If performance collapses, the original raw AUC was likely driven partly by background differences between strata.

Supplementary Note 2: Interpretation of the three AUCs

Raw AUC is computed over all available isolates for the focal organism-drug-site comparison. It is the standard external performance metric.

Matched AUC is computed after removing strata that lack enough resistant and susceptible isolates for the focal drug. It asks whether performance remains in the subset where within-background comparisons are possible.

Background-centered AUC subtracts the mean model score within each valid background stratum before pooling and recomputing AUC. It is the strictest metric because it removes score shifts shared by all isolates in a background stratum.

Supplementary Note 3: How to read low-retention rows

Low retention is not a model failure. It is an audit limitation. If a drug-site pair has few strata containing both resistant and susceptible isolates, then the data do not support a strong within-background conclusion. These rows should be reported as cautionary rather than interpreted as biological evidence.

This is why DRIAMS-B ciprofloxacin is not used as main evidence despite a high centered AUC. The estimate is based on only 25 matched isolates and one valid stratum. The method intentionally makes this visible.

Supplementary Figure 1: Background sensitivity across model families

Fig. 3 | Completed model-class background-audit matrix

Cells show raw AUC -> background-centered AUC for CNN/Mega, LightGBM variants and Weis LR compatibility rows.

Target / site	CNN/Mega	LGBM multi	LGBM single	Weis LR
E. coli / Cipro / A-2018	0.82 -> 0.70	0.83 -> 0.64	0.83 -> 0.65	0.73 -> 0.69
E. coli / Cipro / DRIAMS-C	0.75 -> 0.65	0.81 -> 0.64	0.80 -> 0.68	0.63 -> 0.82*
E. coli / Cipro / DRIAMS-D	0.67 -> 0.60	0.72 -> 0.58	0.71 -> 0.59	0.67 -> 0.62
E. coli / Amox-Clav / A-2018	0.65 -> 0.54	0.63 -> 0.48	0.61 -> 0.47	0.56 -> 0.53
E. coli / Amox-Clav / DRIAMS-C	0.54 -> 0.50	0.59 -> 0.52	0.56 -> 0.53	0.63 -> 0.58
E. coli / Amox-Clav / DRIAMS-D	0.56 -> 0.49	0.58 -> 0.45	0.56 -> 0.49	0.65 -> 0.52
S. aureus / Oxa / A-2018	0.84 -> 0.76	0.79 -> 0.69	0.84 -> 0.73	not run
S. aureus / Oxa / DRIAMS-B	0.78 -> 0.86	0.77 -> 0.81	0.82 -> 0.87	0.67 -> 0.67*
S. aureus / Oxa / DRIAMS-C	0.73 -> 0.70	0.66 -> 0.62	0.72 -> 0.73	0.54 -> 0.54

* low matched support

Figure 1: **Background sensitivity is not specific to the convolutional model.** The same background-matched audit is applied to CNN (top) and multi-task LightGBM (bottom) predictions. Both model families show attenuation from raw to centered AUC, with stronger residual signal for ciprofloxacin than amoxicillin-clavulanic acid across the primary interpretable sites. This indicates that background sensitivity is a property of the data structure, not the architecture.

Supplementary Figure 2: Cross-resistance network

Supplementary Table 1: Claim guardrails

Table 1: Claim guardrails used throughout the manuscript.

Status	Statement
Supported	MALDI-AMR prediction is background-sensitive, and raw AUC can overstate focal-drug signal.
Supported	Background-centered AUC separates retained within-background ranking from background-driven attenuation.
Supported	Public WGS-linked Bruker data show that ST131 lineage is strongly encoded in MALDI peak features.
Not claimed	The DRIAMS CNN is definitively detecting ST131.
Not claimed	The discriminative DRIAMS or UPEC peaks have definitive protein identities.
Not claimed	All MALDI-AMR performance is clonal confounding.
These guardrails distinguish the evaluation-framework claim from direct clone or protein-identification claims.	

Fig. 4 | Co-resistance blocks define exploitable background

Phi correlations across *E. coli* resistance labels show structured resistance ecology.

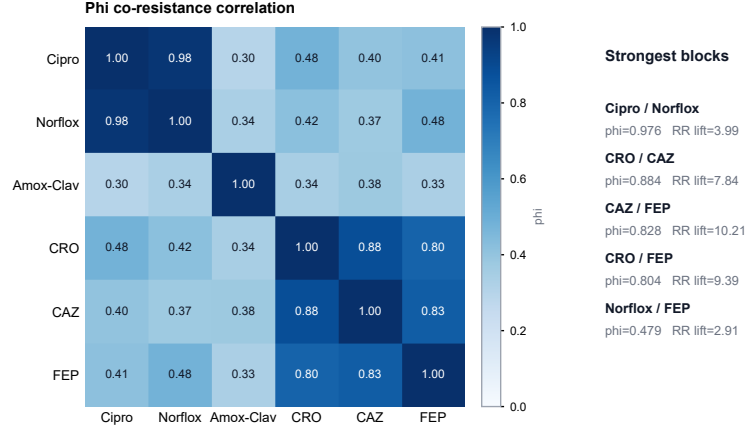


Figure 2: **Cross-resistance network in the expanded *E. coli* panel.** Edge width and shading encode pairwise phi correlation. Two strong blocks are evident: a fluoroquinolone block (ciprofloxacin/norfloxacin, $\phi = 0.976$) and an extended-spectrum cephalosporin block (ceftriaxone, ceftazidime and cefepime, $\phi = 0.804$ – 0.884). These co-resistance structures provide exploitable background signal for any model predicting drugs within either block. Source data: `source_data_fig4_cross_resistance_edges_all_sites.csv`.

Supplementary Table 2: Compact model-family table

Supplementary Table 3: Full *K. pneumoniae* six-drug audit

Supplementary Table 4: Source data files

Supplementary Note 4: What would strengthen the biological claim

The current evidence supports background sensitivity and the need for background-matched evaluation. It does not directly identify the DRIAMS clone or protein features. The next strongest additions would be:

- WGS or MLST labels linked directly to DRIAMS isolates.
- Clone-controlled public WGS-linked MALDI analyses within ST131 and non-ST131 strata.
- Independent MALDI-AMR deployment validation on a non-DRIAMS dataset.
- MS/MS, MALDI-TOF/TOF or LC-MS/MS validation of discriminative m/z peaks.

Table 2: Compact model-family replication of raw-to-background-centered attenuation.

Site	Drug	CNN/Mega	LGBM multi	LGBM single	Weis LR
A-2018	Cipro	0.823->0.703	0.828->0.640	0.828->0.652	0.734->0.689
DRIAMS-C	Cipro	0.750->0.646	0.814->0.637	0.804->0.682	0.625->0.816*
DRIAMS-D	Cipro	0.671->0.596	0.723->0.576	0.712->0.591	0.667->0.620
A-2018	Amox-Clav	0.650->0.541	0.635->0.482	0.614->0.465	0.564->0.529
DRIAMS-C	Amox-Clav	0.535->0.497	0.593->0.518	0.559->0.533	0.626->0.578
DRIAMS-D	Amox-Clav	0.557->0.486	0.580->0.449	0.559->0.487	0.652->0.524

Cells show raw AUC -> background-centered AUC. Asterisks mark caution rows with low matched support. Weis LR cells combine the temporal A-2018 export with existing Weis-style DRIAMS-C/DRIAMS-D compatibility rows; they are compatibility evidence rather than pooled benchmarks with the CNN/LightGBM panel.

Table 3: Full K. pneumoniae six-drug audit results and Weis ceftriaxone extension.

Model	Site	Drug	Raw AUC	Centered AUC	Background AUC	Retention (%)	Strata	Interpretation
CNN/Mega	A-2018	Amox-Clav	0.743	0.532	0.885	79.6	3	background only competitive and weak residual
CNN/Mega	DRIAMS-B	Amox-Clav	0.331	-	0.994	0.0	0	not interpretable or sparse
CNN/Mega	DRIAMS-C	Amox-Clav	0.646	0.473	0.931	38.6	3	background only competitive and weak residual
CNN/Mega	DRIAMS-D	Amox-Clav	0.612	0.484	0.888	93.2	6	background only competitive and weak residual
CNN/Mega	A-2018	FEP	0.896	-	0.999	0.0	0	not interpretable or sparse
CNN/Mega	DRIAMS-B	FEP	0.443	-	1.000	0.0	0	not interpretable or sparse
CNN/Mega	DRIAMS-C	FEP	0.649	-	1.000	0.0	0	not interpretable or sparse
CNN/Mega	DRIAMS-D	FEP	0.647	0.396	0.998	0.9	1	background only competitive and weak residual
CNN/Mega	A-2018	CAZ	0.809	0.208	0.996	2.1	1	background only competitive and weak residual
CNN/Mega	DRIAMS-B	CAZ	0.371	-	1.000	0.0	0	not interpretable or sparse
CNN/Mega	DRIAMS-C	CAZ	0.679	-	0.999	0.0	0	not interpretable or sparse
CNN/Mega	DRIAMS-D	CAZ	0.623	0.929	0.999	0.5	1	cautionary background only competitive but residual signal
CNN/Mega	A-2018	CRO	0.804	0.500	0.981	3.3	1	background only competitive and weak residual
CNN/Mega	DRIAMS-B	CRO	0.404	-	1.000	0.0	0	not interpretable or sparse
CNN/Mega	DRIAMS-C	CRO	0.698	-	1.000	0.0	0	not interpretable or sparse
CNN/Mega	DRIAMS-D	CRO	0.630	0.424	0.973	81.8	2	background only competitive and weak residual
CNN/Mega	A-2018	Cipro	0.788	0.634	0.796	84.6	5	background only competitive but residual signal
CNN/Mega	DRIAMS-B	Cipro	0.578	0.763	0.809	79.1	1	background only competitive but residual signal
CNN/Mega	DRIAMS-C	Cipro	0.702	0.634	0.939	53.7	3	background only competitive but residual signal
CNN/Mega	DRIAMS-D	Cipro	0.587	0.498	0.598	96.4	6	background only competitive and weak residual
CNN/Mega	A-2018	Piperacillin-Tazobactam	0.738	0.531	0.985	7.7	2	background only competitive and weak residual
CNN/Mega	DRIAMS-B	Piperacillin-Tazobactam	0.376	-	0.923	0.0	0	not interpretable or sparse
CNN/Mega	DRIAMS-C	Piperacillin-Tazobactam	0.665	0.477	0.969	15.1	1	background only competitive and weak residual
CNN/Mega	DRIAMS-D	Piperacillin-Tazobactam	0.598	0.526	0.659	95.3	7	background only competitive and weak residual
Weis LR	DRIAMS-B	CRO	0.509	0.509	-	100.0	1	near chance or ambiguous
Weis LR	DRIAMS-C	CRO	0.672	0.672	-	100.0	1	cautionary moderate within background signal
Weis LR	DRIAMS-D	CRO	0.596	0.596	-	100.0	1	weak or partial within background signal

The CNN/Mega rows summarize the K. pneumoniae six-drug panel. The Weis LR ceftriaxone rows are reported separately because they come from the official-panel compatibility export; background-only AUC was not computed for those rows.

Table 4: Source-data files distributed with the manuscript package.

Display item	File	Description
Figure 2	source_data_fig2_primary_background_audit.csv	Primary raw and background-centered AUC values.
Figure 3	source_data_fig5a_public_wgs_maldi_auc.csv	Public WGS-linked MALDI AUC values.
Figure 3	source_data_fig5b_proteomic_biomarker_enrichment.csv	ST131 biomarker enrichment values.
Supp. Fig. 1	source_data_fig3_model_family_replication.csv	CNN and LGBM model-family audit row
Supp. Fig. 2	source_data_fig4_cross_resistance_edges_all_sites.csv	Cross-resistance edges used for the phi h